CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 7: **Measurement of Time and Motion** | Quick revision, one page

Units of time

- SI unit of time: second (s). 1 minute = 60 s; 1 hour = 60 minutes = 3600 s; 1 day = 24 hours.
- Old time-measuring devices: sundial (shadow), water clock, sand clock; modern: quartz clocks and watches.

Oscillation and time period of a pendulum

- Oscillation: one complete to-and-fro motion of the bob back to its starting position.
- Time period $= \dfrac{\text{total time}}{\text{number of oscillations}}$.
- Time period of a simple pendulum depends on its length (longer pendulum, larger period); it does not depend on the mass of the bob.

Speed

- Speed $= \dfrac{\text{distance}}{\text{time}}$; distance = speed $\times$ time; time = distance $\div$ speed.
- SI unit: metre per second (m/s); also km/h. 1 km/h $= \dfrac{5}{18}$ m/s; 1 m/s $= 3.6$ km/h.
- Average speed $= \dfrac{\text{total distance}}{\text{total time}}$.

Types of motion and graphs

- Uniform motion: equal distances in equal times; non-uniform: unequal distances in equal times.
- Periodic motion repeats at regular intervals (pendulum, swing, Earth's rotation).
- Distance-time graph: straight slanting line = uniform speed; horizontal line = at rest; steeper line = greater speed.
- Odometer measures distance travelled; speedometer shows speed of a vehicle.

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